

SMARTHEARING: SCENE-ADAPTIVE HEARING ASSISTANCE VIA LALM-GUIDED LANGUAGE-QUERIED SOUND ENHANCEMENT

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ABSTRACT

Conventional hearing aids prioritize speech, yet which sound matters most—a siren, an alarm, a scream—depends on the acoustic scene. Speech-centric models suppress non-speech warnings, and scene classifiers rely on predefined labels that do not transfer to unseen environments. We propose a scene-adaptive hearing assistance framework that couples a large audio language model (LALM) for open-vocabulary scene understanding with language-queried sound separation. A lightweight Query-space Adapter bridges the representation shift between LALM-generated descriptions and the separator’s training queries, and an adaptive window selects 4–10 s of context per sound category. On 600 mixtures, the Adapter improves SNRi and SI-SDRi by 0.97 dB and 1.14 dB over the base separator; on 145 unseen ESC-50 mixtures, the gains persist (+1.32 dB, +1.19 dB). In a 101-participant listening test, the system was rated most salient in 50.5% of trials versus 28.1% and 21.5% for baselines.

Index Terms— Scene-Adaptive Hearing Assistance, Large Audio Language Models, Speech Enhancement

1. INTRODUCTION

As the population ages and hearing impairment becomes more prevalent, audio assistive devices play an increasingly important role in medical restoration and intelligent health care. Hearing-impaired listeners regularly encounter complex acoustic environments—restaurants, shopping malls, multi-speaker meetings—where background noise, mixed sounds, and overlapping speech coexist. Conventional hearing aids primarily amplify human voices, so other sounds that carry critical information, such as a car horn on the street, may be attenuated, creating fatal risk.

Over the past few decades, audio signal processing has shifted from classical statistical modelling to a deep-learning-driven end-to-end paradigm. Early environmental audio analysis relied on Acoustic Scene Classification (ASC), which assigns a single predefined label—such as “office”—to a



Fig. 1: Everyday scenes where the target to enhance depends on the scenario: (a) a siren on a street, (b) speech in a restaurant, (c) an infant crying at home, (d) a vehicle horn while driving.

recording. Real-world scenes, however, are far more complex, flexible, and scenario-specific than these labels capture. ASC also generalizes poorly beyond its training distribution, limiting its adaptability to unseen environments. Audio processing should therefore move beyond scene classification to identify the relevant acoustic factors and sound patterns that characterize an environment[1], and further reason about how different components interact to anticipate potential danger.

Auditory Scene Analysis (ASA) attempts to model the human auditory attention mechanism, which does not treat all signals equally but prioritizes them according to environmental context, behavioral objectives, and potential danger[2]. Meanwhile, recent Large Audio Language Models (LALMs) such as Audio Flamingo 3[3] and Qwen3-Omni[4] have demonstrated strong audio understanding and causal reasoning[3, 4]. These models provide a basis for an end-to-end architecture that uses LALM reasoning to identify salient acoustic patterns and guide downstream audio enhancement.

This work develops a scene-adaptive sound extraction and enhancement pipeline that adapts to complex environments without per-scene retraining. It uses LALMs for acoustic-component recognition and contextual reasoning, and inte-

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grates their outputs with an audio enhancement model to form a complete hearing-assistance system. This design enables four capabilities: (1) audio-component detection, (2) reasoning about the contextual importance of different sounds, (3) adaptation to unseen acoustic scenarios, and (4) prediction of potential future events. Because the system targets real-world use, robustness and real-time performance are also central to the design.

2. RELATED WORK

Traditional hearing assistance focuses on speech enhancement and noise reduction[5]. Speech enhancement and separation treat the human voice as the sole target, regardless of the acoustic environment[6, 7]. Personalized speech enhancement (PSE) conditions on an enrollment utterance to extract a target speaker, but remains speech-centric and requires re-enrollment for each new target[8]. Non-speech targets appear mainly in music source separation, with predefined categories.

Acoustic scene classification (ASC) assigns a single predefined label to an environment[1], while auditory scene analysis (ASA) models individual functions such as separation or event detection but lacks a unified framework[2]. Scene-aware hearing aids switch between presets based on ASC, which generalizes poorly to unseen or mixed scenes. None of these approaches adaptively determine *which* sound to enhance according to the scene.

Language-queried audio source separation (LASS) extracts a target sound from a natural-language description[9, 10]. CLAP maps text and audio into a shared embedding space, enabling semantic conditioning[11, 12]. However, the same sound can be described in many ways, and LALM-generated queries may not match the distribution seen during LASS training[13]. Recent LALMs exhibit strong audio understanding and reasoning[4, 3], but existing LALM-based hearing-aid works focus on fitting or assessment rather than real-time target selection. To our knowledge, this work is the first to combine LALM-based open-vocabulary scene reasoning with LASS for scene-adaptive hearing assistance, bridging the semantic gap with a Query-space Adapter.

3. METHODOLOGY

3.1. Problem Definition

Traditional hearing aids usually focus on overall enhancement, speech enhancement, or general noise reduction. In real scenes, however, the importance of different sounds depends on their acoustic context and on the sound sources themselves. This raises the central problem of this work: how to perform scene-adaptive audio processing under complex, dynamically changing acoustic scenes. Given a mixed

audio window x_t and no ground-truth scene label, the system must infer the active sound sources and their contextual roles, extract each source, and assign gains $g_{k,t}$ such that safety-related sounds are enhanced, speech and environmental context are preserved, and irrelevant noise is attenuated. The output is

$$y_t = \sum_{k=1}^{K_t} g_{k,t} s_{k,t} + g_{b,t} b_t, \quad (1)$$

where $s_{k,t}$ are the extracted sources and b_t is the residual background. The core challenge is to achieve this with a single model that generalizes across scenes without per-scene retraining or predefined target labels.

3.2. Overall Architecture

As shown in Figure 2, SmartHearing consists of a perception path, an execution path, and an interface connecting them. The input mixture $x(t)$ passes through a ring buffer and is resampled to 16 kHz and 32 kHz, feeding the LALM-based perception path and the LASS-based execution path, respectively. The two paths run asynchronously so that model inference does not block audio processing: the perception path produces a structured scene analysis, including the role, danger level, and confidence of each sound object; the execution path converts the resulting language descriptions into corrected text embeddings, separates the corresponding sources, and recombines them with scene-dependent gains.

3.3. LALM-Based Perception Path

The perception path uses Qwen3.5-Omni to analyze 16 kHz windows. Unlike acoustic scene classification, which outputs a single label, the LALM decomposes the scene into multiple sound objects, each represented as

$$o_k = (\text{sound}_k, \text{role}_k, d_k, p_k, u_k), \quad (2)$$

where sound_k is the sound name, role_k its processing role, d_k its danger level, p_k the recognition confidence, and u_k its relevance to the user. The system uses six roles: urgent warning, other safety sound, direct speech, user-desired content, environmental context, and unwanted noise. This design is not tied to predefined scene labels and can handle unseen scenes.

3.4. Perception–Execution Interface

A frozen CLAP text encoder maps each target description q_k to a 512-dimensional embedding:

$$e_k = E_{\text{CLAP}}(q_k). \quad (3)$$

Since different phrasings of the same sound produce different embeddings, and LALM-generated queries may not

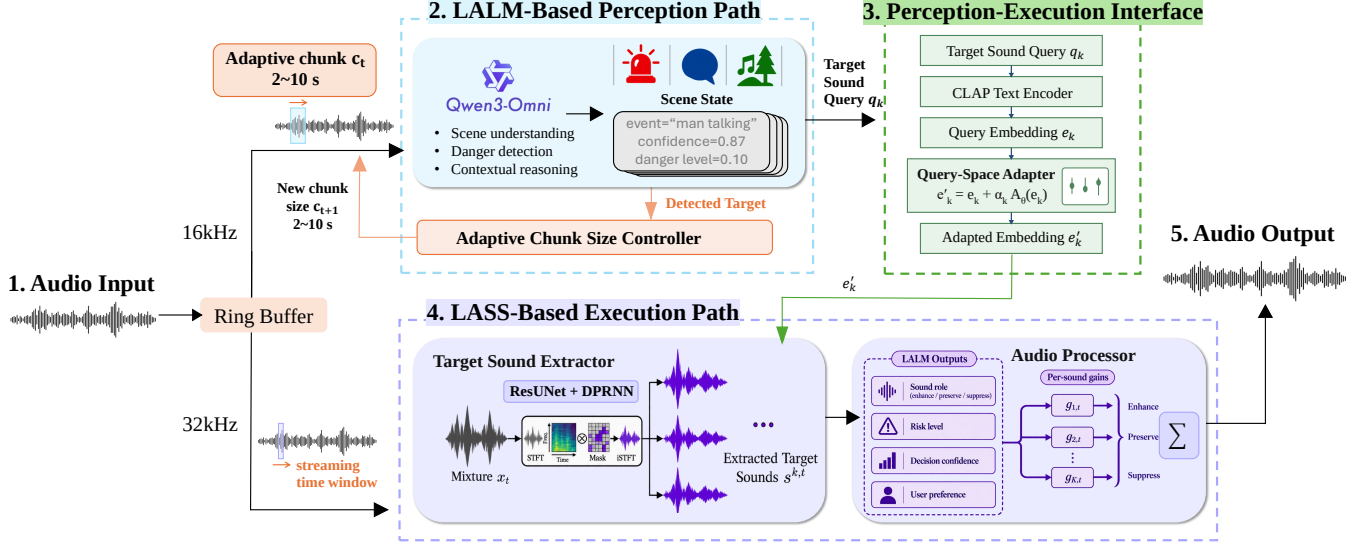


Fig. 2: Overall architecture of the proposed SmartHearing system.

match the distribution seen during LASS training, a Query-space Adapter applies a residual correction:

$$e'_k = e_k + \alpha_k A_\theta(e_k), \quad (4)$$

where A_θ is a residual MLP and α_k controls the correction strength. The last layer is initialized to zero, so $e'_k = e_k$ at the start of training and the Adapter gradually learns a small correction.

3.5. Execution Path

The execution path separates sounds with LASS and then adjusts their gains with an Audio Processor.

Target Sound Extractor Given the corrected embedding $e'_{k,t}$, LASS estimates each target source:

$$\hat{s}_{k,t} = F_{\text{LASS}}(x_t, e'_{k,t}). \quad (5)$$

The system runs LASS multiple times to obtain independently controllable tracks $\hat{S}_t = \{\hat{s}_{1,t}, \dots, \hat{s}_{K_t,t}\}$. Internally, LASS computes an STFT, applies a ResUNet encoder with DPRNN for long-range temporal modeling, and predicts magnitude and phase corrections through a mask-based decoder; we refer the reader to [9] for details.

Audio Processor The Audio Processor assigns gains based on the scene reasoning and the LALM confidence:

$$r_t^{\text{eff}} = r_t (0.5 + 0.5c_t), \quad (6)$$

so that low-confidence decisions lead to more conservative processing. Gain ceilings are calibrated on the validation set and scaled by the effective risk: high-risk sounds approach the validated maximum, while irrelevant sounds receive negative gains. Enhancement, preservation, and suppression are

therefore continuous outcomes of a single gain policy. The final output is

$$y_t = g_{b,t} b_t + \sum_{k=1}^{K_t} g_{k,t} s_{k,t}, \quad (7)$$

where $s_{k,t}$ is the k -th extracted sound, $g_{k,t}$ its gain, b_t the residual background, and $g_{b,t}$ the background gain.

4. EXPERIMENTS

Dataset. Adapter V1 is trained on FSD50K development [17]; V2 adds ESC-50, UrbanSound8K, NIGENS [18], and TUT Rare Sound Events 2017 [19]. Each of the seven target classes contributes 500 training and 100 validation mixtures, with ten interferers and SNRs sampled from -10 to 10 dB.

Baselines. We compare against the Wiener filter[15], Base LASS[9], GTCRN[16]. The LALM perception path is compared under the same frozen LASS separator using Qwen3.5-Omni[4] and Gemini 3.5 Flash[20].

Scene-Adaptive Performance. Table 1 compares SmartHearing against speech denoisers, a target sound extraction model, and non-adaptive baselines. *Oracle Query + LASS* uses the ground-truth target description and serves as an upper bound; “inferred” denotes scene prior derived from audio alone. Speech denoisers collapse on non-speech targets (-18.4 dB SI-SDRi for GTCRN), a fixed query collapses (-22.8 dB), and a recent TSE model[14] underperforms even with ground-truth labels (2.6 dB SNRi), confirming that neither speech-centric priors nor waveform-based TSE transfer to safety-related environmental sounds. A prompt-engineering baseline reaches 10.21 dB SNRi, indistinguishable from Base LASS (10.24 dB), showing that the Adapter’s

Table 1: Scene-adaptive/transfer performance comparison. Values are mean $\pm$ standard deviation across scenes.

Method	Scene Prior	SNRi (dB)	SI-SDRi (dB)	PESQ	STOI
Oracle Query + LASS	✓	12.448 $\pm$ 4.009	11.289 $\pm$ 4.808	2.641 $\pm$ 0.462	0.712 $\pm$ 0.107
Semantic Hearing TSE[14]	✓	2.625 $\pm$ 3.356	-1.434 $\pm$ 4.612	1.562 $\pm$ 0.238	0.531 $\pm$ 0.121
Wiener Filter[15]	–	-0.635 $\pm$ 3.739	-0.751 $\pm$ 3.728	1.478 $\pm$ 0.229	0.566 $\pm$ 0.110
GTCRN[16]	–	-2.415 $\pm$ 3.594	-18.383 $\pm$ 9.741	1.295 $\pm$ 0.199	0.437 $\pm$ 0.110
Fixed Query + LASS	–	-2.553 $\pm$ 4.474	-22.809 $\pm$ 12.830	1.445 $\pm$ 0.312	0.354 $\pm$ 0.152
Prompt Engineering + LASS	–	10.207 $\pm$ 4.575	9.009 $\pm$ 5.340	2.472 $\pm$ 0.480	0.703 $\pm$ 0.104
Qwen3.5-Omni + Base LASS	Inferred	10.239 $\pm$ 4.520	8.989 $\pm$ 5.420	2.516 $\pm$ 0.481	0.702 $\pm$ 0.106
Gemini 3.5 flash + LASS	Inferred	10.370 $\pm$ 4.807	9.109 $\pm$ 5.495	2.548 $\pm$ 0.451	0.699 $\pm$ 0.103
SmartHearing	Inferred	11.241 $\pm$ 4.841	10.317 $\pm$ 5.601	2.603 $\pm$ 0.478	0.710 $\pm$ 0.101

Table 2: Ablation experiment results. STOI differences remain within noise and are omitted.

Comparison	SNRi (dB)	SI-SDRi (dB)	Δ PESQ
V1 – Base	+0.362 [0.155, 0.568]	+0.393 [0.025, 0.761]	+0.033 [0.013, 0.052]
V2 – Base	+0.970 [0.598, 1.341]	+1.138 [0.637, 1.639]	+0.073 [0.036, 0.110]
V2 – V1	+0.608 [0.254, 0.962]	+0.745 [0.274, 1.216]	+0.040 [0.004, 0.076]

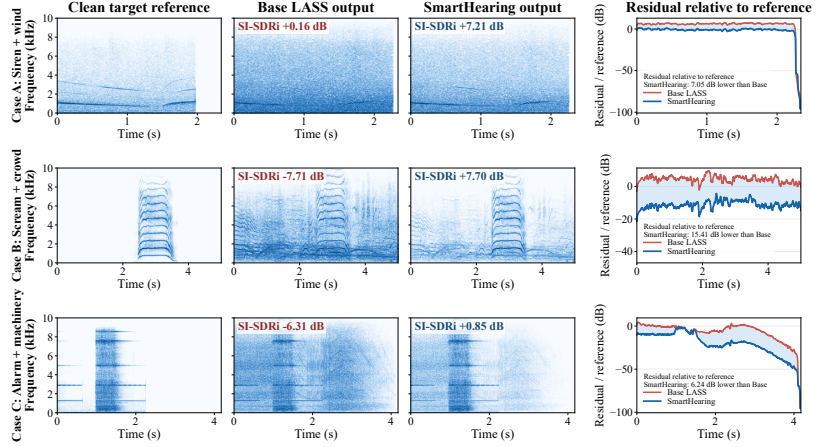
Table 3: Embedding before/after Adapter.

Condition	Within-class $\uparrow$	Between-class	Dist. to LASS $\downarrow$
Raw	0.785	0.337	0.181
Adapter	0.860	0.485	0.119

gain cannot be replicated by query design alone. SmartHearing, receiving no ground-truth label, recovers most of the oracle gap and outperforms an alternative LALM under the same separator. Inference is dominated by the LALM API (median 1.24 s) versus 0.067 s of local separation. A window sweep over ten lengths, six safety sounds, and ten interference types (6,000 units) shows no single window is optimal (Bark 4 s, Siren and Crying 10 s), motivating the adaptive mechanism. Table 2 isolates the Adapter: V1 already improves over Base LASS, and V2, trained on a larger mixture set, significantly outperforms V1 in most metrics.

Embedding Geometry. We compare the geometry of CLAP embeddings before and after the Adapter. Within-class similarity increases from 0.785 to 0.860, showing that paraphrases of the same target become more consistent. More importantly, the distance to the canonical LASS training queries drops from 0.181 to 0.119, confirming that the Adapter moves LALM-generated queries toward the distribution LASS expects. Between-class similarity also rises (0.337 to 0.485), which we attribute to this shared convergence toward the LASS query region rather than a loss of discriminability.

Case Study. Figure 3 gives target reference, scale-aligned output, and residual-energy curves for a siren under wind in-

**Fig. 3:** Case studies: output spectra and residuals relative to reference.

terference. SmartHearing strengthens the siren’s frequency stripes while suppressing background, reducing residual energy by 7.05 dB over Base LASS and raising SI-SDRi from 0.16 dB to 7.21 dB.

Subjective Evaluation. In a questionnaire with 101 responses (303 scene-level choices), SmartHearing was selected as the condition where the target was easiest to notice in 50.5% of trials, versus 28.1% for Base LASS. The mean overall subjective score, however, did not differ significantly across conditions ($p = 0.074$), indicating that increased target salience does not yet translate into overall listening comfort.

5. CONCLUSION

Conventional hearing aids mainly focus on speech enhancement or rely on predefined training scenarios. We propose an adaptive sound enhancement architecture, SmartHearing, to distinguish sounds with diverse levels of importance in complex acoustic scenes. Experiments show that a proper Adapter between perception and execution paths can improve the separation quality of target sounds. Future work will incorporate cross-checking of LALM outputs to improve decision confidence as well as real hearing-impaired evaluation.

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